

The (β) swap: closure, hinge, and the finite $\rightarrow$ limit degeneration

An LLM-attempted derivation — **not verified**

Worktree note

2026-06-10

Status: LLM-ATTEMPTED, NOT VERIFIED. Per CLAUDE.md (“verify LLM proofs independently”) and the verify-by-building discipline, every claim is a candidate for the user to check by hand or in Lean. Confidence flags are the model’s and are fallible. Primary source: `navara_1992.pdf` (PAMS 115, 1992), read directly; formulas quoted from p. 428.

1 The (β) construction, fixed precisely

Chosen per the programme’s anti-smuggler orientation (`genealogy_vision.md`: “watch structure appear without being snuck in”): MO_2 is the *primitive atom*; Navara’s state-engineering scaffold $(\mathcal{T} \times \mathcal{S}$, stateless $\mathcal{S}$) is *discarded*.

Block. $\mathcal{U} := \text{MO}_2$, atoms $a, a^\perp, b, b^\perp$, with $a \wedge b = \mathbf{0}$, $a \not\leq b^\perp$ (i.e. $a \not\perp b$), and complementary pairs $\{a, a^\perp\}, \{b, b^\perp\}$. Bounds $\mathbf{0}, \mathbf{1}$. (*Contrast Navara: $\mathcal{U} = \mathcal{T} \times \mathcal{S}$. We drop the $\mathcal{T} \times$ wrapper.*)

Copies + horizontal sum. For each $C \subseteq M$ (M countable) a copy $\mathcal{U}_C$; $\mathcal{V} :=$ horizontal sum of $\{\mathcal{U}_C : C \subseteq M\}$ (copies glued only at $\mathbf{0}, \mathbf{1}$).

Product + constancy. $\mathcal{W} := \prod_{m \in M} \mathcal{V}$; $\mathcal{L} \subseteq \mathcal{W} =$ all f with Navara’s constancy condition (p. 428, verbatim, with $\mathcal{U}_C$ now an MO_2 -copy):

whenever $f(m) \in \mathcal{U}_C \setminus \{\mathbf{0}, \mathbf{1}\}$ for some $m \in M$, $C \subseteq M$, then $m \in C$ and f is constant on C .

Notation. $v \upharpoonright C =$ the element of $\mathcal{L}$ attaining $v \in \mathcal{U}_C$ on C , $\mathbf{0}$ on $M \setminus C$. Witness: $a_n := a \upharpoonright C_n$, $p := \bigvee_n b \upharpoonright C_n$, for disjoint C_n .

2 Closure step re-derived for MO_2 blocks [the load-bearing check]

Goal. Navara’s proof shows $\mathcal{L}$ closed under countable orthogonal joins (σ -orthocompleteness). The one block-dependent step (p. 428):

“Suppose $f(m) \in \mathcal{U}_C \setminus \{\mathbf{0}, \mathbf{1}\} \dots$ All $f_n(m) \leq_{\mathcal{V}} f(m)$, so belong to $\mathcal{U}_C$. All f_n with $f_n(m) \neq \mathbf{0}$ must be constant on C . There is some f_i with $f_i(m) \neq \mathbf{0}$. For all f_j with $f_j(m) = \mathbf{0}$, the relation $f_i \perp f_j$ implies f_j attains values from $\mathcal{U}_C$ on C . Hence f_j is constant ($= \mathbf{0}$) on C .”

The step that must carry for MO_2 : “ $f_i \perp f_j$ and $f_i(m) \neq \mathbf{0} \Rightarrow f_j$ constant $= \mathbf{0}$ on C .” Navara’s reason: $f_j \leq f_i^\perp$, and the elements $\leq f_i^\perp$ in the block are limited enough to force $f_j = \mathbf{0}$ on C .

Derivation (confidence: MEDIUM-HIGH on the lattice fact, MEDIUM on sufficiency). Take $f_i(m) = x \in \{a, a^\perp, b, b^\perp\}$ (nonzero non-**1**; if $f_i(m) \in \{\mathbf{0}, \mathbf{1}\}$ the constancy clause is not triggered). In MO_2 , for an atom x ,

$$\downarrow(x^\perp) = \{\mathbf{0}, x^\perp\} \quad (x^\perp \text{ an atom; nothing strictly below it}).$$

So $f_j(m) \leq x^\perp$ forces $f_j(m) \in \{\mathbf{0}, x^\perp\}$.

- $f_j(m) = \mathbf{0}$: the desired “= 0 on C ” at coordinate m . ✓
- $f_j(m) = x^\perp$: then f_j is nonzero-non-**1** on the block, so by constancy f_j is constant $= x^\perp$ on C — *not* = 0. So this branch must be excluded by the *orthogonal-family* hypothesis, not by the block alone.

Where it closes. The family $\{f_n\}$ is pairwise orthogonal. If f_i, f_j were both $\neq 0$ on C (values $x, x^\perp$), they are a within-block orthogonal pair (size 2, allowed). A *third* $f_k \neq 0$ on C needs $f_k(m) \leq x^\perp \wedge (x^\perp)^\perp = x^\perp \wedge x = \mathbf{0}$, forcing $\mathbf{0}$. Hence:

Claim 1 (closure survives, MEDIUM confidence). On any block C , at most *two* family members are nonzero (a complementary pair $x, x^\perp$); all others are $\mathbf{0}$ on C . Countable orthogonal joins are computed coordinatewise with ≤ 2 support per block, stay constant on each C , land in $\mathcal{L}$. **σ -orthocompleteness survives the MO_2 swap.**

This *matches* the notes’ assertion (“only elements $\leq a^\perp$ are $\{0, a^\perp\}$ ”) but names the hidden gap: the nonzero branch $f_j = x^\perp$ is killed by the *family cap*, not the block down-set alone.

VERIFY 1: Is “ ≤ 2 nonzero family members per block, rest 0” sufficient for Navara’s coordinatewise-join-lands-in- $\mathcal{L}$ conclusion? I think yes (each is constant on C ; the join of a 2-element within-block orthogonal pair plus disjoint-support tails is constancy-legal), but confirm against his intended argument — he proves = 0 outright, I get it only after invoking pairwise-orthogonality of the whole family.

3 Finite truncation $\mathcal{L}_2^{(N)}$: statefulness lives here [the limit intuition]

Restrict M to N blocks $C_1, \dots, C_N$ (or take the sub-OML generated by the first N block-coordinates): $\mathcal{L}_2^{(N)}$.

Claim 2 (confidence: HIGH). $\mathcal{L}_2^{(N)}$ is finite, non-Boolean (each block carries the live $a \wedge b = \mathbf{0}$, $a \not\leq b$ gap), and **concrete + stateful**: MO_2 is concrete (order-determining 2-valued states); finite products and sub-OMPs of concrete logics are concrete. So $\mathcal{L}_2^{(N)}$ has a separating state space — **proper probability tables exist at every finite stage.**

This is the “statefulness” the limit intuition expects: present and healthy at every finite truncation. The directed family $\{\mathcal{L}_2^{(N)}\}_N$ with inclusions $\mathcal{L}_2^{(N)} \hookrightarrow \mathcal{L}_2^{(N+1)}$ has $\mathcal{L}_2 = \bigcup_N \mathcal{L}_2^{(N)}$ completed under the σ -join. *The question is what survives the colimit + closure.*

VERIFY 2: that $\mathcal{L}_2$ is the σ -orthocompletion of the directed colimit of the $\mathcal{L}_2^{(N)}$ — finite truncations cofinal, closure only adds infinite orthogonal joins. Plausible from constancy; not checked.

4 The hinge + the degeneration map [verdict]

The (★) test on $p = \bigvee_n b \mid C_n$ vs $a_n = a \mid C_n$, computed in $\mathcal{L}_2$ (not $\mathcal{W}$).

Fact (1): $p \wedge a_n = \mathbf{0}$. **(HIGH.)** Coordinatewise: on C_n , $b \wedge a = \mathbf{0}$; off C_n , disjoint support $\rightarrow \mathbf{0}$. Meets are coordinatewise, not deformed by closure (closure adds joins). $\checkmark$

Fact (2): is $p \perp a_n$, i.e. $p \leq a_n^\perp$ in $\mathcal{L}_2$? $\leftarrow$ **THE HINGE**.

Derivation (confidence: MEDIUM — the crux, treat skeptically). $a_n = a$ on C_n , $\mathbf{0}$ elsewhere. Its orthocomplement in $\mathcal{L}$:

$$a_n^\perp = (a \mid C_n)^\perp = a^\perp \mid C_n \vee \mathbf{1} \mid (M \setminus C_n)$$

— $a^\perp$ on C_n , $\mathbf{1}$ everywhere else. *Confidence MEDIUM-HIGH*: standard orthocomplement of a C_n -supported element; crucially (from the source) it is taken *inside the horizontal-sum copy* $\mathcal{U}_{C_n}$, where for the atom a , $a^\perp$ is the MO_2 orthocomplement (an atom). Constancy preserved. So $a_n^\perp \in \mathcal{L}$. $\checkmark$

Test $p \leq a_n^\perp$ coordinatewise:

- Off C_n : $a_n^\perp = \mathbf{1}$, $p \leq \mathbf{1}$. $\checkmark$ no constraint.
- On C_n : $a_n^\perp = a^\perp$, $p = b$. So the hinge reduces to:

$$(\dagger) \quad \text{In } \text{MO}_2 : \text{ is } b \leq a^\perp ?$$

And in MO_2 , $b \leq a^\perp$ is FALSE — the defining meet-zero $\neq$ orthogonal property ($a \not\perp b$ means $b \not\leq a^\perp$).

Claim 3 (hinge YES — ★ HOLDS — conjecture FALSE). CONFIDENCE: LOW-to-MEDIUM. Contradicts the notes' "leans NO." See the adversarial check before believing it. $p \not\leq a_n^\perp$ for every n , so (★) holds in $\mathcal{L}_2$: p a genuine element, $\{a_n\}$ infinite orthogonal, $p \wedge a_n = \mathbf{0}$ yet $p \not\leq a_n^\perp$. $\mathcal{L}_2$ would be a concrete, σ -orthocomplete, non-Boolean OML exercising descent — the sought point-free relational-probability object.

► Adversarial check on Claim 3 — why I do NOT trust it

The derivation is “too easy” and the notes lean the other way. Three places it can break, by priority:

- 1. The orthocomplement formula may be deformed by closure (HIGH suspicion).** I assumed $a_n^\perp = a^\perp \mid C_n \vee \mathbf{1} \mid (M \setminus C_n)$. Navara *warns* “the lattice operations in $\mathcal{L}$ do not coincide with those of $\mathcal{W}$.” If the join $a^\perp \mid C_n \vee \mathbf{1} \mid (M \setminus C_n)$ is enlarged in $\mathcal{L}$'s completion, or if p 's closed form exceeds b -on- $\bigcup C_n$, the on-block comparison changes. *I did not re-derive p as a closed join* — a real gap.
- 2. $p \leq a_n^\perp$ is an ORDER relation in $\mathcal{L}$, and $\leq$ in a completion can hold for coordinatewise-incomparable elements (MEDIUM).** The completion can add order relations: “coordinatewise $b \not\leq a^\perp$ on C_n ” may not survive as “ $p \not\leq a_n^\perp$ in $\mathcal{L}$.” This is exactly the segregation mechanism — Boolean-ization across blocks could force p below $a_n^\perp$ *globally* though it fails *locally*.
- 3. The segregation argument (notes, 2026-06-08) is not addressed above.** Notes: infinite orthogonal families in $\mathcal{L}$ are disjoint-support $\Rightarrow$ Boolean $\Rightarrow$ non-distributivity trapped in finite blocks. Claim 3 asserts p (one element) escapes the trap by being tested *against* the family, not *in* it. **Which is right is the whole open question.** My derivation tacitly assumes the coordinatewise order *is* the $\mathcal{L}$ order; the segregation view says closure changes it. I lean to distrusting Claim 3, precisely because I skipped the closed-join derivation of p (item 1).

The degeneration picture (holds under EITHER verdict)

Regardless of the hinge: finite truncations $\mathcal{L}_2^{(N)}$ are quantum + stateful (§3). The map $\mathcal{L}_2^{(N)} \rightarrow \mathcal{L}_2$ ($N \rightarrow \infty$, completed) is where state-tables and the quantum gap either stay married (Claim 3 / yes) or divorce (segregation / no). This degeneration — *how the proper probability tables of the finite stages relate to the limit* — is the object the “statefulness reappears in a limit” intuition names, worth characterizing either way:

- **yes:** the gap survives to the limit; p is the witness; tables persist.
- **no:** the limit Boolean-izes; tables survive on the skeleton + finite stages; the gap dies in the passage — an explicit picture of σ -additivity destroying non-distributivity (the descent obstruction, made visible).

5 What to verify, in priority order

1. [crux] **Re-derive p as a CLOSED join in $\mathcal{L}$** (not naive b -on- $\bigcup C_n$). Does the σ -orthocompletion enlarge it? The gap that makes me distrust Claim 3.
2. [crux] **Is $\leq$ in $\mathcal{L}$ the coordinatewise order or the completion order?** If completion order, does it force $p \leq a_n^\perp$ globally (segregation)?
3. [closure] **Confirm Claim 1** (“ ≤ 2 nonzero family members per block”) matches Navara’s intended argument.
4. [colimit] **Confirm $\mathcal{L}_2 = \sigma$ -orthocompletion of the directed colimit of $\mathcal{L}_2^{(N)}$.**
5. **Then, and only then:** the hinge verdict + the degeneration map.

Honest bottom line. I derived a *candidate YES* (Claim 3), but my own adversarial check finds the derivation incomplete at exactly the point (closed-join of p , completion-order) where the notes’ “leans NO” would bite. So this worksheet does **not** overturn the leaning verdict — it **localizes the open question to two concrete sub-derivations** (items 1–2), which is the real deliverable. The hinge is now: *does $\mathcal{L}$ ’s completion enlarge p or add the order relation $p \leq a_n^\perp$?*